Diwakar Adhikari

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Education

MSc in Environmental Science, Central Department of Environmental Science (CDES), Tribhuvan University (TU), Nepal, 2021- 2024, **CGPA: 3.85**

Coursework: Ecology, Pollution, Environmental Chemistry, Environmental Management, Geology, Climate Change, Water Resource Management, GIS, Environmental Modelling, R and Python

Specialization: *Water Resource Management, Restoration Ecology*

Thesis: *Assessment of Sediment Load, Transport Mechanisms, and the Impacts Due to Hydropower Plants in the Marsyangdi River Basin.*

Supervisor: Deep Narayan Shah, PhD., Assistant Professor, CDES, TU

BSc in Environmental Science, Tri-Chandra Multiple Campus, TU, Nepal, 2015-2018

Major: *Environmental Science and Chemistry*, **Elective**: *Zoology*

Project Work: *Status of Water Sanitation and Hygiene of Slum Area: A Case of Balkhu, Kathmandu.*

Advisor: Kishor Kumar Maharjan, PhD., Assistant Professor, Tri-Chandra Multiple Campus, TU

Research Interest

- Integrated and/or Urban Water Systems, Governance and Political Economy
- Modelling Water Systems for Climate-Resilient Infrastructure and Adaptation
- Contaminant Fate, Transport, and Treatment

Research Experience

Independent Project, Aug – Dec 2025
Project: *Hydroclimatic variability under Changing Climate in the Marsyangdi River Basin: Implications for water resources management*. | GitHub link (Codes)

Research Fellow, Department of Botany, Tribhuvan University, Oct 2023 – Sept 2024
Project: *Water-soil-food nexus and Sustainable Livelihoods Through the Mainstreaming of Agrobiodiversity Along the Marsyangdi River Basin, Gandaki Province, Nepal*

Research Assistant, Research Center for Applied Science and Technology, Sept 2023 – Apr 2024
Project: *Assessment of Environmental Pollution at Sisdol Landfill Site*

Research Assistant, Kathmandu University, Jun - Jul 2022
Project: *Spring Water Assessment in Karnali River Basin*

Publications

Upadhaya, M., ***Adhikari, D.***, Thapa, L. B., Yadhav, R. K. P., & Pant, R. (2026). Assessment of drinking water quality and irrigation suitability in the Marsyangdi river Basin, Nepal: implications for SDG 6 and ecosystem sustainability (**Accepted**). Environmental Earth Science.

Pant, R. R., Zhang, F., Qaiser, F. U. R., Varol, M., ***Adhikari, D.***, Wang, G., Saqr, A. M., & Bishwakarma, K. (2025). Trace elements in fluvial sediments of the Gandaki River Basin, Central Himalaya, Nepal: Distribution, sources, and risk assessment. *Journal of Soils and Sediments*. doi.org/10.1007/s11368-025-04091-x

Pant, R. R., Koirala, M., ***Adhikari, D.***, & Gaire, N. P. (2024). Ecological Restoration in Nepal: A Comprehensive Policy Overview. In Z. Shang, A. A. Degen, D. Gauchan, M. Koirala, & M. K. Rafiq (Eds.), *Sustainable Ecological Restoration and Conservation in the Hindu Kush Himalayan Region* (pp. 149-166). CABI. doi.org/10.1079/9781800622579.0011

A*dhikari, D.,* Upadhaya, M., Shah, D.N., Thapa, L. B., Yadhav, R. K. P., Srivastava, A, & Pant, R. (**Revision Submitted to Journal of Sedimentary Environments**). *Sediment dynamics and environmental implications of the peaking run-of-river hydropower plants in the Marsyangdi River, Nepal.*

Adhikari, D., Thapa, L. B., Yadhav, R. K. P., & Pant, R. (**Submitted to Hydrological Processes**). *Hydroclimatic variability under climate change in the Marsyangdi River Basin: Implications for water resources management.*

Conference Presentations

Pradhananga, N., Nathaniel, A., Shrestha, A., ****Adhikari, D.,*** Karmacharya A*., Nepal Waste Map,* 4th ISWA-YPG online conference, 2021

****Adhikari, D***., Pradhananga, P. & Karmacharya, A. (2021). *Establishing a community-driven governance system through Digital Waste Management System*. Society of Environmental Engineers of Nepal, Ecocity Satellite Conference, 2021.

**Presenting author*

Scholarships/Grants/Awards

Research Fellowship, Department of Botany, ***Tribhuvan University***, Oct 2023

Data Fellowship, ***Code for Nepal,*** Jul 2023

Most Innovative Development Project (Runner up), ***Global Development Network***, Dec 2021

Climate Action Fund, ***The Asia Foundation***, Dec 2021

Digital Solution Systems for Province 2, ***The Asia Foundation***, Sept 2021

Small Asian Small Grant Project (SASGP), ***The Asia Foundation***, Mar 2021

Technical Skills

Hydrologic modeling: SWAT+, HEC-HMS, watershed delineation and calibration

GIS & Remote Sensing: QGIS, ArcGIS; geospatial data collection and mapping

Programming & Data: Python; statistics and visualization; Excel; basic scientific computing

Climate & Hydrology: GCM selection, climate impacts assessment

Field & Lab: water/soil/sediment/leachate sampling, QA/QC, lab analysis

Professional Experience

Program Coordinator, Nepal Peacebuilding Initiative, Jul 2024 - Jun 2025
Project: *Strengthening Policy-level and Grassroot Capacity to Enhance Climate Change Adaptation for Marginalized Communities*

Consultant, People in Need Nepal, May – Jun 2023
Project: *Urban Resilience Project in Balaju and Godavari*

Program Officer, CLEAN UP NEPAL, Dec 2018- Feb 2022
Various Projects and Programs related to Integrated Digital Waste Management Systems and Air Pollution

Other Activities

- Member of UNESCO Ecohydrology Youth Network (EHYN) since October 2025
- Member of International Water Association – Water Professional Group, Nepal Chapter since 2024
- Term Papers (2023):
  - Irrigation Suitability and Physicochemical Characteristics of Intermittent River: A Case Study of Halche Khola, Nuwakot
  - A Comparative Assessment of Physicochemical Characteristics of Soils of Dry and Moist Regions: A Case study of Manthali Municipality, Ramechhap
- Trained executives of Birgunj Metropolitan City on using digital dashboard and mobile application usage for waste management, Jul 2022
- Did a 38-week-long social media campaign on eco-friendly and sustainable lifestyles titled **Eco-Tips**, Jun 2020 – Jul 2021.

References

1. Deep Narayan Shah | Assistant Professor | CDES | Tribhuvan University
   Email: dnshah@cdes.edu.np
2. Ramesh Raj Pant | Assistant Professor | CDES | Tribhuvan University
   Email: rpant@cdes.edu.np
3. Ram Kailash Prasad Yadav | Professor | Central Department of Botany | Tribhuvan University
   Email: ramkailash.yadav@cdb.tu.edu.np
4. Uddab Raj Khadka | Professor and Head of Department | CDES | Tribhuvan University
   Email: ukhadka@cdes.edu.np
5. Subindra Bogati | Founder and Executive Director | Nepal Peacebuilding Initiative
   Email: subindra@nepalpeacebuilding.org
6. Rabindra Lamichhane | Executive Director | CLEAN UP NEPAL
   Email: Rabindra.lamichhane@cleanupnepal.com.np